Multi-sensory Anti-collision Design for Autonomous Nano-swarm Exploration

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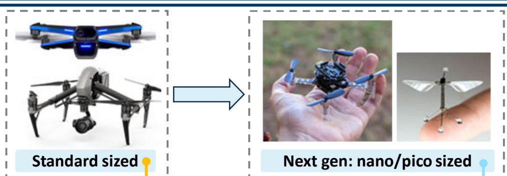
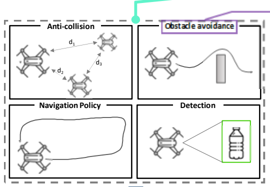
Abstract

This work presents **a robust design for swarm of palm-size nano-drones** enabling autonomous exploration via obstacle avoidance, intra-swarm anti-collision and vision-based target detection capabilities fully aboard a **highly resource-constrained robotic platform at less than 1W power budget**. We combine lightweight single-beam laser ranging to avoid proximity collisions with a long-range vision-based obstacle avoidance deep learning model (i.e., PULP-Dronet) and an ultra-wide-band (UWB) based ranging module to prevent intra-swarm collisions. Our **multi-sensory design** improves collision avoidance with statistic obstacles up to 80% in cluttered environment w.r.t. the state-of-the-art baseline. At the same time, the UWB-based sub-system shows a 92.8% success rate in preventing collisions between drones of a four-agent fleet.

Nano-drone Challenge



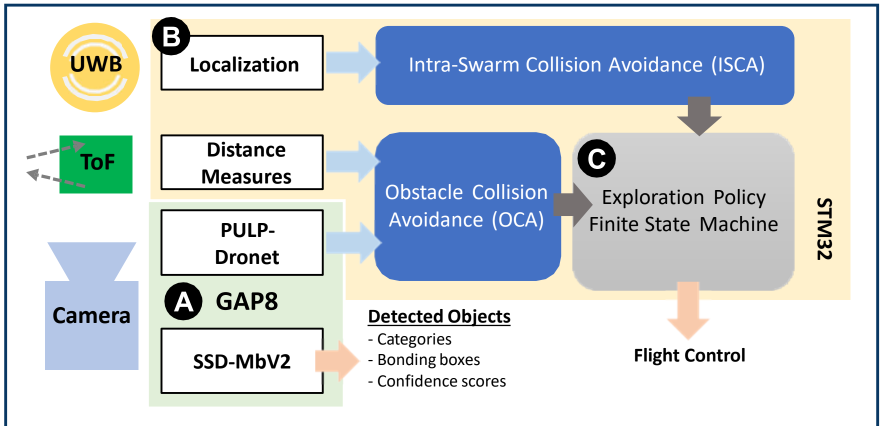
	Standard-sized	Next-generation
Size [ø, weight]	~50cm / ~ few Kg	~few cm / ~few g
Tot. Power	~ 100 W	~ 5W
Processing device	High-end CPU	Low-power MCU
Cognitive Capacity	Fully autonomous	



Laser-based obstacle avoidance
Fast but ineffective in cluttered env.
Vision-based obstacle detection
Effective but slow high computation

Need for a highly efficient and compact design suitable for nano-drone swarm [Architectures & Algorithms]

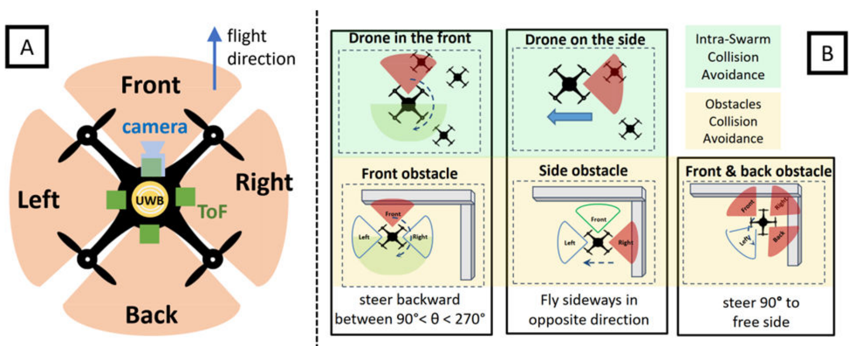
System Design



Ⓐ **RISC-V Dual-CNN architecture at 250kB Memory** is a precise embedding of two deep-learning base algorithms handling obstacle detection and object detection in an interleaving process.

Ⓑ **UWB-based anchorless localization** enables swarm agents to coordinate each other relatively and avoid colliding with each other during exploration

Ⓒ **Sensor fusion and exploration policy** continuously monitor sensing inputs and react accordingly based on the sensor-fusion algorithm.



[A] System setup [B] Exploration policy based on the multi-sensory collision avoidance logic.

Results

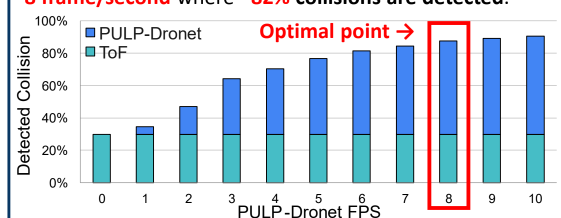
► Proposed system is evaluated based on obstacle avoidance ability, optimal PULP-Dronet throughput and intra-swarm collision detection followed by system costs.

► **Multi-modal sensing achieves 60%-70% boost in obstacle avoidance** where beam-sensing frequently fails

Env.	Parameters	SniffyBug	ToF-only	ToF& PULP-Dronet
obstacle free	Crash-free rounds	5 /5	5 /5	-
	Avg. crash/min	0	0	-
	Avg. coverage/min[%]	20.7	22.4	-
obstacle populated	Crash-free rounds	4 /20	3 /20	16 /20
	Avg. crash/min	1.8	1.5	0.2
	Avg. coverage/min[%]	33.2	22.6	10.3
narrow corridor	Crash-free rounds	0 /10	1 /10	7 /10
	Avg. crash/min	2.6	2.4	0.4
	Avg. coverage/min[%]	19.8	45.2	26.7

↑60% ↑70%

► PULP-Dronet delivers **optimal obstacles detection at 8 frame/second** where **~82% collisions are detected**.



► **Multi-sensory system** performs at **>1W power budget**

► **Intra-swarm anti-collision** unit achieves **92.8% accuracy**

► **Dual-CNN design** operates at **250kB memory & 133mW power** where GAP8 processor executes at **1.6FPS**

MCU	Task	Throughput [Hz]	Parameters [MB]	Memory [KB]	Power [mW]
STM32	UWB	20	-	0.4	320.1
	ToF	20	-	0.1	286 [6]
GAP8	SSD	1.7	4.67	250	134 [6]
	PULP-Dronet	62.9	0.0832	200.4	100.9
	SSD&PULP-Dronet	1.6	4.7532	250	133.1

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